

Assignment 2 - Verifying an agentic GIS routing workflow

Submission Date: Sunday, September 27th, 2026

Class presentations: Monday, September 28th, 2026

Think back to the fiber route planning task from Week 3. The agent may have produced code, a map, a route comparison and other outputs, but how do we know whether these results are correct and defensible? Use the framework from 04_01 and the verification and validation exercises in 04_02 to investigate this question.

Create a 5-minute presentation that explains the routing agent approach, presents evidence from your own checks, and identifies which conclusions you trust, and which remain unresolved. Focus on the existing shortened corridor analysis. You do not need to develop a new agent application, but you should show your results, along with other supporting materials developed for verification/validation.

Begin by explaining the six framework dimensions in Table 1. Keep this introduction brief so that most of your presentation is devoted to verification techniques/findings. Identify one decision that should remain with a human and one example of feedback that should change the agent's next action.

Table 1 Framework aspects to consider.

Framework component	Question to address
Goal	What outcome was the routing agent asked to produce?
Environment	Which saved inputs and settings could it observe?
Tools	Which GIS operations and code libraries did it use?
Agency	Which decisions could it make, and which limits and judgments were set by a human?
Feedback	What evidence should lead it to revise its next action or stop?
Verification	What checks would establish that its computations meet the specification? What requires independent evidence?

Use the saved Week 3 inputs and outputs, the original settings, and AGENTS.md. Preserve the exact endpoints of route1_short.gpkg, with the easterly endpoint as the start; measure in EPSG:32737 and retain the original study rectangle. Compare baseline, roads, power, and rail using the same positive cost per meter. Preserve source data and the first comparison.

Write down at least five expected checks before asking an agent to audit the results. Cover endpoints and continuity, valid geometry and containment, meter-based length, connector limits and counting, and cost arithmetic. Recompute quantities from the saved evidence rather than accepting the agent's report. Use the 0.01 m tolerances specified in AGENTS.md where applicable. Record the expected result, observed result, method, evidence, and PASS, FAIL, or NOT_CHECKED for each check. Missing evidence is not a pass.

Review the five synthetic routing cases required in AGENTS.md and explain one in your presentation, including its expected outcome. Show that a connector is counted once. If endpoints remain disconnected after permitted repairs, the correct result is no_path with blank length and cost; it is not a fabricated straight-line route.

Complete the gap tolerance sensitivity exercise in 04_02, so reduce the saved gap tolerance from 50 m to 25 m, or use a smaller positive value if your original setting differs. Change only that setting and save the rerun separately. Compare route availability, length, connector use, and the cheapest available candidate. An unchanged result is still useful evidence.

Distinguish computational verification from real-world validation. Identify two possible independent evidence sources and explain their provenance and limitations. Inspect one location using supplied or approved evidence if available; otherwise state what remains unresolved. Discuss the simplified assumption that 2D crossings connect and why mapped infrastructure does not establish access rights or construction feasibility.

Please submit your presentation online as a PDF by September 27th. You will present it to the class on Monday, September 28th. Aim for approximately five main slides: problem and agent framework; route map; verification evidence; sensitivity and validation; conclusions and proposed improvements. Include source acknowledgments and a brief AI-use statement. Supporting slides may contain the full check table and details that will not fit within five minutes.

Retain your audit script, run command, package versions, check table, and separate sensitivity results under the output folders specified in 04_02, and identify these evidence files in your supporting slides. The submitted presentation should include at least one route map with inferred connectors distinguished and OpenStreetMap attribution, a readable summary of your checks, and the sensitivity comparison.

The emphasis is on the quality of your investigation and your explanation of the evidence. Finding and documenting an error or a justified no_path result is valuable. A successful agent run alone does not demonstrate a successful analysis. Table 2 outlines the assessment criteria.

Table 2 Assessment criteria.

Criterion	Evidence	Weight
Agent framework	Clearly explains the six dimensions, human responsibilities, and feedback or stopping conditions.	20%
Verification evidence	Uses expected and observed results, reproducible checks, and a synthetic case to assess the routing approach.	35%
Sensitivity and validation	Compares the controlled rerun and assesses independent evidence, spatial assumptions, and unresolved claims.	25%
Communication	Clear, legible presentation within five minutes, with an informative map, justified conclusions, sources, and AI-use statement.	20%

Finally, students may use AI tools in accordance with the course AI policy. Yet, students remain responsible for understanding, testing, validating, and documenting all submitted material. The presentation must include a brief AI-use statement identifying the tools used and explaining how they contributed. Undisclosed AI-generated material, fabricated citations and other materials, or code that the student cannot explain may constitute an Honor Code violation and will be reported.